Nutri-Scan Technical Pack

Technical readiness, deployment and field operations documents

Prepared from the saved Nutri-Scan workspace documentation on 15 October 2026.

Document status: planning and review drafts. This PDF is not commercial, legal, scientific, regulatory or production approval.

Generated locally from Markdown and CSV source files. Refer to the source files for ongoing edits and the claims matrix before external use.

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README

Nutri-Scan technical and operations launch pack

Prepared against the working tree on **2026-10-15**. This is an operational planning pack, **not a launch approval or evidence that application checks ran**. All checklist boxes start unverified. Assign named owners and attach dated evidence for the exact release, environment and device before changing their status.

Current product boundary

- The API uses an explainable, **uncalibrated prototype scoring engine**. Scores,

risk bands and confidence are screening indicators, not laboratory results, calibrated probabilities or feed-safety certification.

- All ten screening inputs are required. Numeric NIR/vision inputs are adapters;

uploaded photos are private evidence, not automatic image analysis. No validated protein/fiber prediction or contaminant/mycotoxin detection is established.

- Farm accounts use SQLite and HttpOnly sessions with owner/member/viewer roles.

This is an MVP; a broad production multi-tenant release requires further review.

- The ESP32 sketch defaults to **synthetic demonstration readings**, USB serial,

unknown calibration and a manually fixed timestamp. Real sensors require validated integrations. Serial and voice entry propose values for explicit review only.

- Offline drafts work in an already-loaded signed-in app and need explicit sync.

The production PWA's offline reload provides **public help only**, not farm history, private attachments or authenticated offline workflows. Drafts discard instrument provenance. Testing requests and reminders are farm-local records, not lab bookings or SMS services.

Documents and operating order

Document	Use / accountable role
Technical readiness checklist	Establish scope, owners and evidence; release lead
Deployment runbook	Dev, staging, HTTPS deployment, backup, migration and rollback; operations
Hardware integration and wiring	Serial contract, electrical review and device records; hardware lead
QA test plan	Regression, acceptance, target-device and field pilot gates; QA lead
Calibration and model release	Sample/lab provenance, leakage controls and release criteria; calibration lead
Data governance and privacy	Access, retention, deletion, backup and incident operations; data steward
Support and incident runbook	Triage, containment, recovery and escalation; support/on-call lead

Document	Use / accountable role
Risk register	Product, technical, safety and business decisions; release lead
Field operator SOP	Preflight, sample handling, capture, review and handover; field lead
Production launch checklist	Final go/no-go and monitored handover; release authority

Start with readiness, rehearse deployment and QA in staging, resolve the risk register, then complete the final launch checklist. Use the field SOP for every pilot visit and the incident runbook when evidence or safety is in doubt.

CSV templates

These are **manual operational records**, not API import files or the calibration CLI's input schema. Each has a header and one `placeholder_not_for_use` row; remove that row in a controlled operational copy before recording real activity. Blank timestamps/measurements mean **not recorded**, never zero, passed or calibrated.

Template	Row meaning and key
calibration-record.csv	One reference point/check per device sensor; `calibration_record_id` + `point_id`
device-inventory.csv	One physical board/instrument; stable `device_id`
qa-test-results.csv	One execution of one test; `run_id` + `test_case_id`
incident-log.csv	One incident summary, with timeline in restricted evidence; `incident_id`
sample-chain-of-custody.csv	One collection/transfer/receipt/analysis event; `sample_id` + `event_id`

Record conventions

- UTF-8, comma-delimited, one header; quote text containing commas/newlines/quotes.

Preserve headers when copying. Treat spreadsheet formula-like text as text rather than executing it. No passwords, cookies, keys, raw personal contacts or full farm reports in these repository templates.

- `data_status`: `placeholder_not_for_use`, `synthetic_demo_not_for_field_use`, or

`real_field_record`. Real status describes origin, **not validation or approval**. Keep synthetic and real collections in separate controlled files/releases.

- `*_at` and `due_at`: timezone-aware ISO 8601; calendar dates: `YYYY-MM-DD`.

`*_pct`: percent; `temperature_c`: degrees Celsius. Record wet/dry basis explicitly. Numeric cells are finite decimal numbers or blank; do not use `NaN`, `inf` or `0` to fill a missing measurement. SHA-256 fields hold the lowercase 64-digit digest of the referenced artifact when available.

- IDs and `*_ref` fields point to restricted records; use pseudonymous farm/operator

IDs. Evidence references must resolve for reviewers without making files public.

- QA `status`: `not_run|pass|fail|blocked|not_applicable`; a pass requires actual

results/evidence and executor/time. Calibration `result`: `pending|pass|fail`; protocol calibration states remain `unknown|reported_current|reported_expired`.

- Inventory `mode`: `demo|real_pending_validation|validated_for_defined_use`;

inventory `status`: `not_in_service|in_service|quarantined|retired`.
Incident `status`: `open|contained|monitoring|closed`; severity follows the incident runbook. Custody `event_type`: `collection|transfer|receipt|preparation|analysis|disposal`.

Source of truth and reconciliation

Read these existing sources alongside the pack:

1. Project README: current scope, accounts and normal commands.
2. Same-day demo: emulator/ESP32 demonstration and its limits.
3. Fieldwork verification: **previously reported**

checks and remaining real-device gaps, not a result from this documentation task.

1. Fieldwork contract: private routes, attachment

limits, observation/request state machines and current PWA boundary.

1. ESP32 README: serial and sensor hooks.
2. Calibration workflow: synthetic scaffold and

requirements for real reference data.

1. Backend operations, architecture

and farmer usability: supporting context.

Some earlier architecture/usability text predates the fieldwork PWA extension.
For installation/offline behavior use the current fieldwork contract and PWA implementation: public offline help exists; private offline reload does not.
The README's legacy analyze route is authenticated today; the default ESP32 sketch does not POST to it. Deployment records the current runtime configuration, including always-present localhost origins and SameSite=Lax limits.

Evidence and placeholders

`PLACEHOLDER_*`, `.example.invalid` hostnames and sample service/path names are operator substitutions, never existing infrastructure or credentials. Commands are instructions to run in the stated directory with pre-existing dependencies; their inclusion does not mean they were executed. No default account, API key, model registry, automated retention job or production service unit is supplied.

Record a release ID, source revision **and working-tree/artifact digest**, backend and frontend artifact digests, configuration revision (redacted), device/firmware versions, QA results and named approvals in a restricted release evidence location. A Git revision alone cannot identify a build made with uncommitted changes.

Calibration And Model Release

Calibration and model release control

The application currently reports `kind=prototype_screening`, `model_version=prototype-baseline-0.1`, `calibration_status=uncalibrated` in its assessment model. Its engine is a transparent formula, not a trained field model. Device calibration, model calibration and qualification of advisory language are **separate approvals**.

Source procedures: calibration workflow, demo runbook, hardware integration.

1. Lifecycle and accountable owners

Stage	Required record / owner	Promotion gate
Intended use	Sample types, population, outputs/units/basis, exclusions and claims; product/domain lead	Qualified expert agrees the measurement contract
Device qualification	Wiring, firmware/time source, sensor reference checks; hardware lead	Defined-use electrical and repeatability criteria pass
Collection design	Sampling plan, farm/batch/time grouping, lab methods and thresholds; calibration lead	Protocol frozen before examining performance
Collection + reference	Raw instrument data, custody events, lab reports/QC; field/lab leads	Traceable matched sample IDs and acceptable handling/QC
Dataset freeze	Version, manifest/hash, exclusions, split assignment; data steward	Reviewed provenance and leakage audit; synthetic excluded
Fit + validate	Train-only preprocessing/fit, validation decisions and reproducible artifacts; model lead	Predeclared criteria and independent review
Locked test	Held-out farms/batches/time, uncertainty and subgroup results; independent reviewer	Test opened only after design/model frozen; all approved thresholds met
Release	Model card, runtime integration tests, compatibility and rollback; release authority	Signed scope-specific approval and staged QA
Monitor / retire	Drift, reference rechecks, incidents, replacement/withdrawal record; model/calibration leads	Trigger-based revalidation and controlled retirement

No automatic model loader, artifact registry or runtime coefficient import is provided by the current scaffold. Promoting a future model requires an explicitly implemented and tested application change; a JSON coefficient file alone changes no application behavior.

2. Synthetic scaffold — exact demonstration commands

From the root, use an existing private temporary output directory. These commands are instructions, not claimed results from this pack:

```
ls -ld /tmp/omnirush
export CAL_DEMO_DIR="$(mktemp -d /tmp/omnirush/nutriscan-calibration.XXXXXX)"
python3 -m unittest discover -s tools/calibration -p 'test_*.py' -v
python3 tools/calibration/calibrate.py \
  --input tools/calibration/sample_reference.csv \
  --output "$CAL_DEMO_DIR/calibration-demo.json" \
  --fit-calibration
python3 -m json.tool "$CAL_DEMO_DIR/calibration-demo.json"
```

The CLI checks headers/row shape/IDs/finite numbers/duplicates, deterministically splits whole farm groups using SHA-256 ranking, reports baseline MAE/RMSE/signed bias/R^2, and optionally fits one univariate line per target on **train only**.
Fitting needs at least three train pairs and nonconstant input for a target.
Validation/test metrics are separate. The sample CSV has six **synthetic** farm groups; it is not an adequate real evaluation dataset.

Every CLI report is labelled `synthetic_demo_not_for_field_use`, even if another input file is supplied. Do not remove that marker, rebrand the scaffold as field validation or paste coefficients into the backend. Keep real evaluation in a controlled, separately approved workflow with true provenance and review.

The CLI's input header is exactly:

```
sample_id,farm_id,sample_type,moisture_sensor,moisture_lab,protein_nir,protein_lab,fiber_nir,fiber_lab,ph_sensor,ph_lab
```

The new operational calibration record
and custody record are **not**
drop-in inputs to that CLI. Maintain reviewed sample-ID joins and a documented transform if exporting paired measurements to an evaluation dataset.

3. Provenance and laboratory reference

1. Assign immutable sample IDs at collection. Record pseudonymous farm, batch/lot,

sample type/variety, date/time/timezone, operator, storage/handling conditions, device/sensor IDs, firmware, preprocessing and sampling-plan version.

1. Use the qualified sampling plan for representative subsamples, homogenization,

clean containers, preservation, transport time/temperature and disposal. Record whether the field measurement and lab aliquot used the same homogenized material. Do not invent universal holding times: the selected lab method specifies them.

1. Preserve exact raw ADC/spectral/other output before viewing reference results,

with immutable evidence reference and SHA-256. Record repeated readings rather than selecting whichever reading later agrees best with the lab.

1. Log each custody event in sample-chain-of-custody.csv:

transfer parties by ID, seal/container condition, time, storage temperature, handling, received lab ID and related report reference. Do not join by row order.

1. Obtain results from a qualified lab using a documented validated method suitable

for the sample/target. Capture method/version, units and wet/dry basis, receipt, preparation/analysis time, instrument/QC run, replicate results, uncertainty and detection/quantification limits where relevant, analyst/run and report checksum.

1. Predefine exclusions for failed QC, insufficient/contaminated samples, broken

custody, unknown basis or impossible values. Quarantine such records for expert review, retain reasons/corrections and do not silently convert them to ground truth.

1. Resolve unit/basis conversions once in versioned preprocessing. Obtain permission

for collection/use, minimize identifiers and follow governance.
Keep a restricted ID mapping separate from analysis exports.

Sensor calibration records additionally capture reference/buffer lots/certificates, raw counts/voltage, environmental temperature, as-found/as-left values, tolerances, due time and reviewer. Plan pre/post-session checks and assess samples since the last known good check if a sensor drifts or fails.

4. Leakage prevention and locked evaluation

- Freeze dataset hashes and a split manifest **before fitting**. Keep farms together

at minimum; strengthen groups for shared batches/lots, storage stacks, device sessions, near-time sampling and related sources. Use later-time evaluation where intended deployment faces drift.

- Keep duplicate aliquots, repeat instrument scans and lab replicates in one group.

Audit cross-split duplicates/raw hashes and relationships, not just sample IDs.

- Fit scalers, imputation, feature selection, preprocessing and calibration on train

only. Choose hyperparameters/thresholds on validation; do not use test labels to select features, exclusions, acceptable error or model form.

- Lock final model/transform/criteria before opening test results. A test-driven

redesign needs a new independent test set; do not relabel the same test as unseen.

- Preserve collection chronology, access history, exclusions, corrections and all

failed experiments in reviewable controlled records. Synthetic examples, lab-result derived features and post-result manual edits cannot enter the real evaluation.

5. Model card and numeric release gates

Create a controlled card for each scoring/model/advisory release. Required fields:

Section	Required contents
Identity	Release ID/version, artifact SHA-256, source/worktree digest, training/evaluation code and dependency versions, owner/reviewer/date
Intended use	Feed/silage domain, regions/varieties/devices/population, user decision, prohibited uses and human escalation
Inputs/outputs	Required measurements, units/basis/ranges, preparation/preprocessing, missing-input handling, runtime compatibility
Data	Collection period, sample/farm/batch counts, provenance/permission, lab methods/QC/uncertainty, dataset and split hashes, exclusions
Evaluation	Per-target/split/subgroup coverage, missingness, MAE/RMSE/bias/R^2, confidence intervals, repeatability, held-out farm/time performance
Decision behavior	Sensitivity/specificity/false negative/false positive rates at approved thresholds where relevant; uncertainty coverage, abstention and out-of-domain behavior
Limitations	Known failure modes, unsupported contaminants/claims, device/season/language limitations and operator requirements

Section	Required contents
Operations	Monitoring/stop triggers, reference-check cadence, artifact/config compatibility, rollback version and owner, retirement/revalidation plan
Approval	Lab/domain, hardware, data, QA, safety/product and release-authority decisions with evidence refs

Before evaluating the locked test, fill and approve a gate record with **numeric** limits per target/subgroup: minimum independent farms/batches/samples and coverage, maximum MAE/RMSE/absolute bias, acceptable uncertainty, repeatability, and maximum false-negative rate at relevant decision thresholds. These limits are currently **PLACEHOLDER_UNSET** and must be set by qualified domain/lab reviewers, not invented from the synthetic metrics. Unset or unmet required gates mean **no model promotion**. A good aggregate R^2 or higher heuristic confidence is insufficient.

6. Staged release, monitoring and withdrawal

Run shadow evaluation against lab/reference results first, then a bounded authorized pilot with predefined sample coverage/stop rules, then only the approved use. Retain the prior runtime/configuration/artifact and run QA, including old record readback and report/model-version labeling. Historical saved results are not silently recomputed on model change; any reanalysis must be a separately traceable record/workflow.

Monitor device drift, out-of-range/missing inputs, subgroup error, reference failures, operator overrides, safety complaints and unsupported use. Thresholds, sampling cadence and responsible reviewer must be named in the card. On a breach, suspend affected use/claims, identify affected device/time/model/sample ranges, preserve evidence, refer to a qualified lab/adviser and follow the incident runbook. A new sensor, firmware conversion, lab method, season/domain expansion or preprocessing change requires impact review and appropriate recalibration/revalidation, not a version-label-only release.

Data Governance And Privacy

Data governance and privacy operations

This is an **operational policy proposal, not legal advice**. The data steward and release authority must approve scope, retention periods, access and local obligations before real-data launch. No scheduled retention, deletion service or audit-log system is implemented by this document. Current behavior comes from the README, backend README and fieldwork contract.

1. Data map and minimization

Location / category	Current contents and handling	Operational owner
SQLite accounts	Username, farm identity, role, scrypt password hash; session token digests/expiry (not plaintext tokens)	Identity/data steward
SQLite screening/history	Farm/batch/sample IDs, input readings, outputs/advisory, timestamps, request IDs, optional instrument provenance	Farm data steward
SQLite fieldwork	Batch metadata, observation answers/notes, reminders, testing-request contacts/status/dates	Field/data lead
SQLite attachments	Private normalized photo BLOBs and lab report/photo BLOBs; request PDFs are downloaded, not OCR-interpreted	Data/operations lead
Browser local drafts	Complete readings scoped by user/farm, queue/request metadata; no password, token, photo or API response; instrument metadata stripped	Device user/field lead
Browser Cache Storage	Public help/manifest/icons/allowlisted assets only; not private farm data or personalized HTML	Frontend/operations lead
Exports and sharing	Downloaded HTML, browser PDF, QR batch routes, optional previewed summaries/native photo sharing	Exporting user/farm owner
Optional voice	Browser speech service may process microphone input externally/network-dependently; Nutri-Scan does not save audio	Device user/data steward
Host logs/backups/evidence	Operator-configured logs, DB snapshots, calibration/custody/incident/QA records	Operations/data steward

Keep contact details out of batch notes unless needed for the chosen testing provider. Use pseudonymous farm/operator IDs in calibration analysis and operational CSVs; store re-identification mappings in a separate restricted location. Avoid faces, addresses, number plates and unrelated material in photos. Photo metadata stripping does not remove identifying content visible in pixels or document text.

QRs contain a route, not a token or access grant, but may expose a batch identifier. Reports include farm details; preview before sharing. WhatsApp/native share is an explicit user action, not automatic background transmission. Once exported to another device/recipient, files are outside server-side access controls; include them in handling/deletion instructions and avoid uncontrolled support attachments.

2. Access and identity operations

- Owners add member/viewer accounts; owners and members can write scans/evidence/

fieldwork, viewers are read-only. Farm scope comes from the session, not client claims. Verify foreign-resource isolation in QA.

- There is no default account. Give each operator an individual account; prefer

viewer access for review. Verify a farm owner's identity before adding users or assigning legacy data. Document grants, purpose, approver and review/expiry time.

- Proposed review cadence: at pilot onboarding, staff changes and monthly while

active. Host DB/backup operators are privileged across farms; keep that group small, use protected host access and record all administrative data access externally.

- Session cookie: HttpOnly, SameSite=Lax, seven days; Secure requires explicit HTTPS

configuration. Password/session hashes remain sensitive and do not make a stolen DB anonymous. Runtime/backup credentials belong in protected operator-managed secret storage, never Git, frontend `VITE_*`, logs or CSV evidence.

- Member removal, account disable/reset/recovery and farm erasure are not complete

self-service administrative workflows in this MVP. Resolve the operational process for a closed pilot and block broader exposure until approved controls exist. Logging out or deleting a session does not disable future credential-based login.

- Scrub request bodies, `Cookie`, `Set-Cookie`, credentials, attachment bytes and

identifying query strings from logs/support bundles. Minimize proxy URL logging; literal history filters can contain farm/batch information. No audit-log completeness is asserted for the app; maintain a restricted operations/change log separately.

3. Proposed retention schedule — approval required

These are concrete **starting defaults**, not current automatic behavior or legal requirements. Enter approved values, owner and purge job/manual procedure in the release evidence before capture. Document exceptions and expiry; do not silently extend retention because a job was never scheduled.

Category	Proposed operational period / trigger	Disposal / verification
Synthetic staging/demo data	14 days after demonstration/test completion	Destroy isolated DB/profile and test exports; retain minimized QA summary
Real pilot scans, fieldwork, photos/reports	Pilot closure + 90 days unless a documented purpose requires another period	Reviewed scoped erasure/export process; verify parent and attachment removal
Account/member data	Active pilot access; review/remove within 30 days of authorized closure/withdrawal	Apply approved account/farm administrative process; revoke sessions, verify scope
Browser drafts/queue	End-of-shift reconciliation; remove confirmed/unneeded copies within 7 days	User checks history before deleting; resolve queue/conflict first; confirm device storage
Downloaded/shared exports	Task completion, target within 7 days unless explicitly retained	User/recipient handles downloads, print copies, share apps and device backups
Redacted host access/error logs	30 days	Restricted log rotation and verified expiry; incident hold only by approval

Category	Proposed operational period / trigger	Disposal / verification
Incident/change/release/QA summaries	12 months; raw private evidence separately minimized	Keep needed non-identifying outcomes; retire raw evidence by category-specific expiry
Real calibration/custody/raw lab evidence	Model/measurement release lifetime + 12 months	Preserve reproducibility and withdrawals; controlled de-identification/erasure review
Encrypted DB backups	Daily points for 30 days, then expire	Verify off-host expiry/replicas and record deletion-ledger reconciliation on restore

An incident/evaluation hold needs a named authority, reason, restricted scope and review date; track it in the same retention ledger. Do not conceal a pending deletion or promise erasure while a documented exception remains active.

4. Backup, restore and deletion reconciliation

Proposed minimum service objectives for approval: **RPO 24 hours**, **RTO 4 hours**, daily consistent DB backups plus a pre-change backup, encrypted off-host copy, daily backup failure review and monthly restore drill. More frequent capture may require a shorter RPO. These are targets, **not measured service guarantees**.

Use the SQLite backup/restore commands. Record timestamp, checksum, size, integrity/FK check, encryption/storage reference, release pairing, retention expiry and drill result. Restrict directory/file access; HTTPS and mode 0600 do not encrypt a DB at rest. Prove that a restore includes attachments, legacy visibility, roles and lazy fieldwork tables. A backup is not qualified until a restore and authenticated readback have succeeded in isolation.

Keep a restricted **deletion ledger** outside restorable DB snapshots: request ID, authorized scope, opaque farm/user/resource IDs needed to reapply deletion, backups containing the data, hold/expiry, executor/reviewer, completion and verification. This ledger itself requires access limits and retention. On restore, revoke old sessions and apply approved erasures **before reopening access**; otherwise a backup can resurrect deleted records. Restore-time reconciliation is mandatory even when the app rollback succeeds technically.

5. Manual deletion/withdrawal workflow

There is no farm-erasure HTTP endpoint or generic purge command. Do not improvise ad-hoc live SQL or assume deleting one farm row cascades all data.

1. Verify requester identity and authority without asking for their password/cookie.

Log opaque request ID, scope (account, farm, scan, attachment, exports or dataset), requested action and acknowledgment date.

1. Inventory affected accounts/sessions, scans/photos, batches/observations/reminders,

testing contacts/reports, legacy-assigned data, local drafts, logs, exports, datasets/model artifacts and all backup copies. Confirm scope with the owner; identify approved holds and downstream recipients.

1. Data steward approves a reviewed, schema-aware administrative procedure in a

maintenance window. Take the protected pre-operation backup under its approved retention/hold, rehearse on a copy, define row/attachment counts and test foreign keys/remaining farms. If a reliable procedure is unavailable, mark the request pending and escalate; never mark it completed merely by hiding data in the UI.

1. Restrict access/revoke sessions as needed, execute the reviewed scoped procedure,

verify affected bytes/records are gone and other farms unchanged. A logical delete alone does not prove physical secure erasure from SQLite pages, journals, SSDs or filesystem snapshots. Select an approved storage-level erasure/rewrite method and verify it in isolation.

1. Guide the authorized user through draft/download/shared-device cleanup after

resolving unsynced work. Logout clears displayed scope, **not necessarily stored drafts or downloads**. Notify identified recipients through the approved process; record limitations without claiming remote recall.

1. Track encrypted backup expiry or controlled destruction and prevent resurrection

through the deletion ledger. Assess dataset withdrawal and any need to retire or retrain affected models; do not promise removal from an artifact by deleting CSV rows.

1. Record executor/reviewer, actual completion/remaining copies, expiry and user-facing

response. Retain only the minimized evidence necessary to prove the operation.

6. Incident and operational checklist

Suspected exposure, lost device, wrong-recipient share, backup theft or cross-farm access triggers the incident runbook. Contain the affected route/device/account, preserve minimal restricted evidence, identify data categories/time/farms/recipients, and assign the incident/data leads. Determine any external communications or obligations with the responsible organization; this pack does not invent a legal notification deadline.

- ☐ Named data steward, access approvers and recovery/deletion operators recorded.
- ☐ Data map and chosen retention schedule approved; actual purge/rotation duties assigned.
- ☐ Registration/access withdrawal/session revocation gaps resolved for scope.
- ☐ HTTPS/Secure cookie/origin/isolation and private cache checks executed.
- ☐ Device-local storage, exported reports, sharing and speech-provider handling explained.
- ☐ Backup encryption, failure alerting, restore drill and deletion ledger tested.
- ☐ Scoped deletion procedure rehearsed, including attachments/backups/other-farm checks.
- ☐ Incident contacts, redaction procedure and restricted evidence access verified.

Deployment Runbook

Deployment, backup, migration and rollback runbook

Operational recipe; not evidence of a deployment. Use an approved release and pre-existing Python/backend environment and Node/frontend dependencies. Provisioning, service units, TLS certificates, backup encryption and alerting are operator-owned; they are not included in this repository. Existing start/check commands are in the README and backend README.

1. Environment contract

Environment	Network / cookies	Storage / use
Development	localhost 5173 → Vite proxy → 8000; secure cookie false only for local HTTP	Disposable dev DB; no real farm data
Staging	Separate HTTPS hostname, same-origin `/api` proxy, secure cookie true	Separate DB, generated accounts and synthetic fixtures; production-like restore/migration rehearsal
Production / restricted pilot	Approved HTTPS hostname, same-origin `/api` proxy preferred, secure cookie true; backend loopback/private only	Durable absolute DB path, encrypted backups, named operators and approved release scope

Use different databases, accounts, backup locations and configuration for each. Production paths/domains below are **placeholders**. Do not copy farm data into staging without an approved minimized/pseudonymized dataset and retention plan.

Variable	Current meaning
`NUTRI_SCAN_DB`	Backend SQLite file; default is `backend/data/nutri_scan.sqlite3`. Relative overrides resolve from process working directory, so deploy with an absolute path.
`NUTRI_SCAN_ORIGINS`	Comma-separated explicit origins (scheme + host + optional port, no paths/wildcards). Localhost/127.0.0.1 ports 5173/5174 are **always added**; this variable does not replace them. Mutations also accept the API's own origin.
`NUTRI_SCAN_COOKIE_SECURE`	Backend `true` for HTTPS; default false. Session cookie is `nutri_scan_session`, HttpOnly, SameSite=Lax, path `/`, seven-day lifetime.
`VITE_PUBLIC_API_URL`	Public build-time API origin for a separately hosted frontend; use empty for same-origin. Legacy fallback is `VITE_API_URL`. Both must be empty for an unambiguous same-origin build.
`VITE_PROXY_TARGET`	Local Vite dev/preview proxy target; not a production reverse proxy or secret.

Backend variables must be supplied by the shell/supervisor; Uvicorn is not configured here to load a root `.env` automatically. Vite build-time variables are public and embedded into assets. Never place passwords, tokens, TLS keys or backup encryption keys in `VITE_*`, the frontend bundle, firmware or release evidence.

Separate origins: a same-site HTTPS API subdomain may work with credentialed requests and the exact frontend origin configured. Truly cross-site hosting can fail because cookies are SameSite=Lax and browser policy applies. Do not assume CORS changes

that cookie rule. Prove register/login/reload/upload/logout on target browsers or use the same-origin topology. Always-present dev origins need a recorded threat review and appropriate edge/network restrictions before wider exposure.

2. Local development

From `backend/`, using its existing virtual environment (Terminal A):

```
NUTRI_SCAN_COOKIE_SECURE=false .venv/bin/python -m uvicorn app.main:app --host 127.0.0.1 --port 8000
```

This uses the default local DB unless explicitly overridden. From `frontend/` (Terminal B), override legacy URL settings to retain the cookie-friendly proxy:

```
VITE_PUBLIC_API_URL= VITE_API_URL= VITE_PROXY_TARGET=http://127.0.0.1:8000 npm run dev -- --host 127.0.0.1
```

Open `http://localhost:5173`. `/api` and `/health` proxy to the backend. Run `docker compose up --build` from the repository root only for the documented development stack: it runs dependency installation and Vite dev at startup, exposes ports and bind-mounts code. It is not the production recipe. No Compose/install command was executed to prepare this pack.

3. Candidate build and staging

Run the QA commands in an isolated checkout/workspace, preserving existing local edits. From `frontend/`, for same-origin hosting:

```
VITE_PUBLIC_API_URL= VITE_API_URL= npm run build
```

Retain the complete `frontend/dist/`, including hashed assets, `sw.js`, manifest, icons and offline help. Record artifact digests and build environment. PWA generation requires Vite's production mode and deployment at **root** `/`; subpath hosting is not supported by the current plugin. Rehearse with a local production preview:

```
VITE_PUBLIC_API_URL= VITE_API_URL= VITE_PROXY_TARGET=http://127.0.0.1:8123 npm run preview -- --host 127.0.0.1 --port 5174 --
strictPort
```

The matching isolated API/DB setup is in QA. For actual staging, serve the candidate static artifact behind its own HTTPS reverse proxy and use a separate absolute `NUTRI_SCAN_DB`; do not use Vite preview as the staging/production service supervisor. Test the same configuration and first authenticated fieldwork request that production will use.

4. HTTPS production topology and startup

Substitute an approved domain, static release path, TLS files and service account. Example backend process command from the deployed `backend/` directory:

```
NUTRI_SCAN_DB=/srv/nutriscan-data/PLACEHOLDER_PRODUCTION.sqlite3 \
NUTRI_SCAN_ORIGINS=https://nutriscan.example.invalid \
NUTRI_SCAN_COOKIE_SECURE=true \
.venv/bin/python -m uvicorn app.main:app --host 127.0.0.1 --port 8000 \
  --proxy-headers --forwarded-allow-ips=127.0.0.1
```

An operator-provided supervisor must supply the same environment, working directory, restart policy and restricted user. Use one API process initially; SQLite write contention and process-local login throttling are not solved by adding workers. Restrict registration/exposure at the edge for an approved closed pilot; there is no built-in invitation-only deployment switch. Review recovery/removal/admin gaps before a public multi-tenant launch.

Illustrative Nginx server blocks (configure on the host, not in this docs pack):

```
server {
    listen 80;
    server_name nutriscan.example.invalid;
    return 301 https://$host$request_uri;
}
server {
    listen 443 ssl;
    server_name nutriscan.example.invalid;
    ssl_certificate /etc/PLACEHOLDER_TLS/fullchain.pem;
    ssl_certificate_key /etc/PLACEHOLDER_TLS/private-key.pem;
    root /srv/nutriscan/releases/PLACEHOLDER_RELEASE/frontend/dist;
    client_max_body_size 11m;

    location /api/ {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header Origin $http_origin;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_cache off;
    }
    location = /health {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_cache off;
    }
    location = /sw.js {
        add_header Cache-Control "no-cache" always;
        try_files $uri =404;
    }
    location /assets/ {
        add_header Cache-Control "public, max-age=31536000, immutable" always;
        try_files $uri =404;
    }
    location / {
        add_header Cache-Control "no-cache" always;
        try_files $uri $uri/ /index.html;
    }
}
```

Serve `.webmanifest` as `application/manifest+json` (or `application/json`), JS/CSS with correct types, and `offline.html` as `text/html`. Verify host MIME mappings. Do not rewrite `/api/*` errors to SPA HTML. Do not cache private requests at a CDN; preserve API `Cache-Control: no-store`, `nosniff` and `Set-Cookie`. Keep unknown Hosts rejected by the host's default vhost. Trust forwarded headers only from the actual proxy, and prevent direct public access to port 8000. Add HSTS only after the HTTPS/certificate/subdomain rollout has been rehearsed.

Read-only edge probes after setting the real HTTPS origin:

```
export APP_ORIGIN=https://nutriscan.example.invalid
curl --fail --show-error --silent "$APP_ORIGIN/health"
curl --show-error --silent -D - -o /dev/null "$APP_ORIGIN/api/v1/auth/me"
curl --fail --show-error --silent -I "$APP_ORIGIN/sw.js"
curl --fail --show-error --silent -I "$APP_ORIGIN/manifest.webmanifest"
```

Expected: health 200 with `status: ok`; unauthenticated `auth/me` 401 with `no-store` / `nosniff`; static responses 200 with appropriate types. `/health` is static service status, not a DB readiness probe. Complete authenticated QA in the browser, inspecting cookie flags without exporting cookie values into evidence.

5. Consistent SQLite backup

Back up before startup/migration, legacy assignment, model/app release or deletion. The DB includes account/session hashes and photo/report BLOBs: protect the whole backup as private farm data. Do not copy only a live `.sqlite3` file with ordinary `cp`; use SQLite's backup API to include committed state consistently.

Set **absolute, operator-provisioned** paths. `BACKUP_DIR` must already exist, be owner-only and be on approved encrypted storage outside web roots/repository. Confirm the parent directories, then use a unique output filename:

```
export NUTRI_SCAN_DB=/srv/nutriscan-data/PLACEHOLDER_PRODUCTION.sqlite3
export BACKUP_DIR=/srv/nutriscan-backups/PLACEHOLDER_PRIVATE_DIRECTORY
ls -ld "$BACKUP_DIR" "$(dirname "$NUTRI_SCAN_DB")"
export BACKUP_PATH="$BACKUP_DIR/pre-release-$(date -u +%Y%m%dT%H%M%SZ).sqlite3"
python3 - "$NUTRI_SCAN_DB" "$BACKUP_PATH" <<'PY'
import os, sqlite3, sys
from pathlib import Path
source, target = map(lambda p: Path(p).resolve(), sys.argv[1:])
assert source.is_file() and source != target
assert target.parent.is_dir() and target.parent.stat().st_mode & 0o077 == 0
fd = os.open(target, os.O_CREAT | os.O_EXCL | os.O_WRONLY, 0o600)
os.close(fd)
src = sqlite3.connect(source.as_uri() + '?mode=ro', uri=True)
dst = sqlite3.connect(target)
try:
    src.backup(dst)
    integrity = dst.execute('PRAGMA integrity_check').fetchall()
    foreign_keys = dst.execute('PRAGMA foreign_key_check').fetchall()
    assert integrity == [('ok',)] and not foreign_keys, (integrity, foreign_keys)
finally:
    dst.close()
    src.close()
print('Backup verified:', target)
PY
sha256sum "$BACKUP_PATH"
```

If this fails, mark the artifact failed and investigate; a leftover file is not a verified backup. Record checksum, timestamp, size, release/schema pairing, encryption location, retention expiry and restore evidence in the restricted backup log. The commands set private file permissions but **do not encrypt**; host storage encryption and encrypted off-host transfer must already be configured and tested.

6. Migration and restore rehearsal

There is no versioned migration runner or down-migration command. Backend startup initializes tables and adds missing scan columns; fieldwork tables initialize lazily on first fieldwork access. Rehearse both against a **copy**, then compare counts, attachments, legacy visibility, roles, old scan/report readback and idempotency. Do not infer backward compatibility from “additive”. Record the actual schema and code version together; never hand-edit production schema as an improvised migration.

Restore a verified backup into a **new**, unused file on an existing private test directory. Stop any service that could use the restore destination. Substitute real approved paths; this leaves the source backup and live DB untouched:

```
export RESTORE_DB=/srv/nutriscan-restore/PLACEHOLDER_UNIQUE_RESTORE.sqlite3
ls -ld "$(dirname "$RESTORE_DB")"
python3 - "$BACKUP_PATH" "$RESTORE_DB" <<'PY'
import os, sqlite3, sys
from pathlib import Path
source, target = map(lambda p: Path(p).resolve(), sys.argv[1:])
assert source.is_file() and source != target
assert target.parent.is_dir() and target.parent.stat().st_mode & 0o077 == 0
fd = os.open(target, os.O_CREAT | os.O_EXCL | os.O_WRONLY, 0o600)
os.close(fd)
src = sqlite3.connect(source.as_uri() + '?mode=ro', uri=True)
dst = sqlite3.connect(target)
try:
    src.backup(dst)
    assert dst.execute('PRAGMA integrity_check').fetchall() == [('ok',)]
    assert not dst.execute('PRAGMA foreign_key_check').fetchall()
    # Restored historical sessions must not become valid again.
    dst.execute('DELETE FROM sessions')
    dst.commit()
finally:
    dst.close()
    src.close()
print('Restored with sessions revoked:', target)
PY
```

Reconcile the deletion ledger **before** restored data is exposed. From the selected release's `backend/`, use the restored DB on an unused loopback port:

```
NUTRI_SCAN_DB="$RESTORE_DB" NUTRI_SCAN_COOKIE_SECURE=false \
.venv/bin/python -m uvicorn app.main:app --host 127.0.0.1 --port 8123
```

Use the isolated frontend/QA procedure, generate no real outbound shares, and record actual recovery time and recoverable data age. Stop rehearsal processes afterward. For real cutover pause writes at the edge, drain/stop the old API, take a final verified backup, then start the candidate and complete authenticated fieldwork smoke tests before reopening writes.

Legacy assignment is a separate **explicit data-ownership change**, never automatic: after reviewing source legacy farm, destination owner and backup, from `backend/`:

```
NUTRI_SCAN_DB=/srv/nutriscan-data/PLACEHOLDER_PRODUCTION.sqlite3 \
.venv/bin/python -m app.admin assign-legacy \
  --username PLACEHOLDER_EXISTING_OWNER --farm-id PLACEHOLDER_UNOWNED_LEGACY_FARM
```

Verify exact scan IDs and attachments under the intended farm and that other farms cannot read them. Record authorization and before/after counts without raw data.

7. PWA deployment, update and rollback

- Publish the complete static release atomically; retain the previous artifact and

its hashed assets for existing tabs. For a new release, carry forward older hashed assets from the approved artifact store during the transition, or use an equivalent immutable asset route. A new HTML file with missing old chunks breaks open tabs.

- Serve the new `sw.js` with revalidation. Worker cache identity includes public

assets and policy. Updates **wait for explicit user activation**; do not force reload while readings are unsaved. On activation, older namespace caches are removed.

- Inspect Cache Storage: only allowlisted static assets/help/manifest/icons, never

API/auth/history/photos/reports or personalized `index.html`. Public offline help must remain public and carry no account details.

- Test online load, update notification, explicit update, offline reload and all

three help languages on HTTPS target devices. Installation availability depends on browser policy; manual browser installation may be required.

8. Rollback and recovery

Trigger on cross-farm exposure, unsafe claims, corruption/data loss, broken login, failed migration or required acceptance failures. Incident lead owns the decision.

1. Restrict access/pause writes; announce impact and preserve redacted logs plus a

verified post-failure DB snapshot. Record the last known good release and time.

1. If schema/data compatibility was proven in staging, stop the candidate, restore

the prior backend and full frontend artifact/configuration, retain the current DB, then run readback and write smoke tests before reopening.

1. If compatibility is unknown or data changed incompatibly, keep service closed.

Restore the pre-change backup into a new DB as above, invalidate sessions, reconcile deletions and obtain approval for the known recovery-point data loss. Reconcile post-backup operations from preserved evidence; do not blindly replay writes or offline queues. Same farm/request ID with changed content returns 409.

1. Switch the supervisor's DB path/release pointer while stopped, then restart under

the old approved code. Supervisor commands depend on the host's actual unit; no universal `systemctl` unit or deployment script is supplied here.

1. Redeploy the **old worker and its assets together**, revalidate `sw.js`, and guide

users through explicit update after saving/handling drafts. Do not indiscriminately clear all site storage: it can destroy unsynced drafts. Verify cache/privacy again.

1. Re-run scoped QA, verify counts/attachment downloads, monitor, communicate recovery

and complete the incident review. Document any unrecoverable scans, permissions changes or sample provenance gaps.

Use governance for backup/deletion controls and production launch checklist for go/no-go evidence.

Field Operator Sop

Field operator standard operating procedure

Use for an **authorized pilot or clearly labelled synthetic demonstration**.
The field lead supplies the approved sampling method, qualified adviser/lab contacts and support route before travel. Nutri-Scan is an uncalibrated screening prototype; neither a low-risk score nor a photo establishes that feed/silage is safe.

Source workflows: same-day demo,
fieldwork contract,
farmer usability and
hardware instructions.

1. Preflight and start-of-shift record

- Confirm scope (demo versus real pilot), site permission, operator training and

support/qualified adviser availability. Record date/time/timezone, release, operator ID and weather/storage conditions as needed by the sampling plan.

- Confirm phone/browser/USB support, charged equipment, reachable HTTPS app and

correct individual account/farm. Owner/member can submit; viewer can only read. Do not share accounts or store passwords in notes/templates.

- Inspect electronics, leads, enclosure and clean sample-contact surfaces. Do not

modify powered wiring, attach raw pH probes or put a 5 V output on ESP32 ADC.

- Check device ID, firmware/mode, real time source and calibration/reference records

against inventory. Perform the approved reference check and record it in calibration-record.csv.

- If reference checks fail, calibration is unverified/expired, equipment is damaged

or capture time is stale, stop using that instrument for real measurements. Record the issue and use a qualified alternative or observation-only workflow.

- For a demo, label devices, batch IDs and every associated file **synthetic demo —

not field/lab measurements**. Default ESP32/emulator numbers cannot populate real farm records. The serial wire has no demo flag, so the external label is essential.

2. Identify the batch and take a representative sample

1. In **Farm fieldwork**, create/select the exact feed/silage batch. Record human

batch ID, name/location and known preparation/opening dates/quantity. Do not guess dates, invent expiry or convert quantity units silently. Batch identity (ID + sample type) is immutable after creation; fix errors through the field lead.

1. Assign immutable sample ID(s), including aliquot relationships, **before**

measurement. Follow the qualified plan for representative subsamples, mixing, clean containers, preservation/transport limits and protective equipment. Do not taste or deliberately inhale suspect material.

1. Record collection and each handover/receipt/preparation/analysis event in

sample-chain-of-custody.csv.
Include timezone-aware time, sample/batch/device/operator IDs, seal/condition, storage temperature/handling and restricted evidence reference. Keep the actual ID mapping private. Unknown conditions remain explicitly unknown/blank.

1. Clean equipment between samples per method to prevent carryover. If handling,

contamination or custody is uncertain, quarantine the record/sample for qualified review; never “correct” readings after seeing a lab report.

3. Choose observation or full screening

Observation only: choose **Quick observations**, answer visible mould, unusual heat, unusual smell and cover damage as Yes/No/Unknown. Leave unknown as Unknown, save and confirm **Observation only — no screening score**. Add private evidence if authorized. Concerning/unknown observations require qualified follow-up, not an invented numeric diagnosis.

Full screening: Use this batch copies identity only, not previous readings.
The four steps are sample → batch → readings → review/submit. All ten inputs are required; incomplete readings cannot be submitted or saved as complete-reading drafts:

Input group	What to enter
Storage/sensor	Moisture %, pH, temperature °C, humidity % from suitable measured methods
Nutrient	Protein %, dry matter %, crude fiber % from a suitable report/instrument with known basis
Visual prototype inputs	`mold_score`, `color_anomaly`, `surface_damage`, each 0–1 from the trained operator's approved scoring method; a photo does not generate these automatically

Use **How to measure…** and language/listening help, but do not replace the approved method with a guessed value. Missing is not zero. If any required actual value or validated scoring method is unavailable, use observation-only tracking and arrange qualified measurement/testing instead of filling demo values.

4. Optional instrument/voice capture

1. For a validated serial device, close the serial monitor, connect safely and open

Read from instrument (USB serial) on reading step 3. Supported Chromium needs HTTPS/localhost, user permission, the documented JSON protocol and 115200 baud.

1. Explicitly choose the port, review device/time/calibration/reference/units/values,

select only trustworthy readings, acknowledge review and confirm. Calibration status is reported by the device, not verified by the app.

1. On timeout, denied permission, unplug or invalid output, record the failure and

continue with a measured manual alternative. The app neither commands the device nor submits a scan automatically. A new capture replaces prior provenance; edits to captured values remove their provenance.

1. Voice is optional and requires microphone consent. Review every proposed value;

browser speech may use a network/provider service. Nutri-Scan does not save audio. Typed entry remains available. Listening/recognition on real devices can differ from software mocks and draft translated safety guidance needs expert review.

5. Final review, save and interpretation

- Compare every value/unit/basis to the actual measurement record, plus batch/sample,

capture time/source and any device provenance. Correct mistakes **before** final confirmation; edits invalidate the prior confirmation.

- Submit once when online. Wait for the saved result/scan ID, then refresh history

and confirm readback. A spinner/error is not proof that the scan was saved; inspect history before repeating an uncertain submission. Record sample-to-scan linkage in the controlled field record.

- Read saved advisories and prototype status. Never certify safety or change feeding

decisions solely because of score/confidence. If spoilage, unusual heat/smell, suspect material or conflicting lab results are reported, stop using the affected screening as assurance and contact the qualified adviser/lab under the site plan.

- Photos are optional private evidence, not diagnosis: up to five JPEG/PNG/WebP,

each <=5 MiB. Avoid identifying bystanders/locations and upload only with authority.

6. Offline, drafts and retry

- In an **already-open signed-in app**, complete actual readings, choose **Save

draft**, then** Queue** if an upload is intended. Unsaved edits are not saved. Reconnection alone does not upload; press **Sync now** explicitly while signed in with permission for the same farm.

- Check draft/queue counts, last confirmed queued upload and saved history. Failed

uploads keep actionable state; a 409 needs history/support review before changes. Do not create a new request ID just to force an ambiguous queued item through.

- **Drafts retain readings but discard instrument device/time/calibration provenance.**

Keep an independent protected measurement/custody log. If provenance is required, use direct online capture/submission under the approved plan or escalate; never claim draft sync retained it. Incomplete readings need a protected field record, not invented values to satisfy draft validation.

- An offline reload of the installed PWA shows **public help only**. Farm history,

fieldwork writes, reports/photos and authenticated reload need reconnection. Do not clear browser/site storage or accept disruptive updates with unresolved local work. Follow the field lead's approved backup/reconciliation procedure.

7. Follow-up, reports and sharing

- Create an in-app reminder and check it manually; it sends no SMS or background

alert. Complete/reopen it as actual work progresses.

- For testing help, contact the chosen qualified provider **outside the app**,

confirm method/handling/time/price yourself and record only necessary contacts. The saved request is local tracking, not a booked lab service. Mark collected with the actual date; upload a trusted report (PDF <=10 MiB or supported photo, max five) before marking report received. Do not treat basic PDF validation as malware scanning or an uploaded report as automatically interpreted by Nutri-Scan.

- Download saved-result HTML; use browser Print/Save PDF if needed. Reports include

farm identifiers and reproduce saved readings, not later form edits. Review before sharing. Use the preview-first summary and verify recipient/identity/photo choices; identity/photos are off by default. Text actions do not send private photo URLs.

- Batch QR routes require sign-in and grant no access. A localhost QR cannot serve

another phone; use the approved reachable HTTPS deployment.

8. End-of-shift handover and stop conditions

Reconcile samples, custody events, scans, device checks, drafts/queues, reminders, testing requests and lab handovers with the field lead. Confirm every purported saved record by readback, note unresolved items and retain only approved protected evidence. Perform any required post-session reference check; a failure triggers review of samples since the last known good check. Clean/store equipment per method.

Resolve/retain unsynced work before deleting drafts. Log out, secure the device, protect downloads/print copies and follow retention/privacy; logout alone does not erase local drafts or exported files. Record device returns, remaining battery/damage and next calibration/referral tasks.

Stop and escalate unsafe wiring, failed reference checks, unknown sample identity, privacy exposure, data loss or potentially harmful interpretation through support/incidents. Record time, scope and observed facts; do not ask the farmer to treat a prototype score as confirmation of safety.

Hardware Integration And Wiring

Hardware integration and wiring

Current source: ESP32 README,
sketch,
same-day demo, and
instrument validator.
This document does not certify a wiring design or a sensor's accuracy.

1. Demo mode versus real measurement

Item	Current implementation	Required before real use
`DEMO_MODE`	`true`; illustrative 62% moisture, pH 4.2, 24°C, 68% humidity	Label the entire session synthetic; replace/validate hooks before disabling demo mode
Moisture	ADC hook with placeholder dry/wet counts	Material-specific reference method, measured transfer function and repeatability
pH	ADC hook with placeholder voltage/slope	Conditioned probe interface, appropriate sample preparation, buffer calibration and temperature handling
Temperature/humidity	Non-demo hooks return `NAN`	Actual validated DHT/BME-like library/module integration and error handling
Clock	Fixed `MANUAL_CAPTURED_AT`; no RTC/network sync	Validated real capture-time source for real sessions, including timezone and drift check
Nutrient values	Not measured by this sketch	Suitable validated report/NIR instrument; never derive protein/fiber from demo readings
Network	No automatic Wi-Fi, credentials, POST or serial commands	A separately reviewed integration if networking is introduced later

Non-finite/out-of-range hook results are omitted. The browser requires at least one valid reading and the screening still requires all ten inputs. A plausible emitted number does not make the placeholder conversion calibrated. Even after updating a manual timestamp before a demo build, repeated frames carry the same timestamp; that is not continuous real measurement time.

2. Electrical preflight — hardware lead sign-off required

Disconnect power before changing wiring. Identify the **exact board variant**, module supply/output limits, ADC configuration and interface circuit from datasheets. The following pins are editable placeholders, not universal ESP32 pin assignments:

Signal	Sketch default	Integration caution
Conditioned pH analog output	GPIO 35	**Never connect a raw pH probe or a 5 V output directly to ADC.** Use the probe manufacturer's high-input-impedance signal-conditioning/interface circuit and verify its output range.

Signal	Sketch default	Integration caution
Moisture analog output	GPIO 34	Verify material suitability, common ground and worst-case signal voltage; supply voltage is not the same as safe output voltage.
DHT-like data	GPIO 4	Placeholder only; review pull-ups/logic voltage and board boot-strapping constraints; no measurement library is present.
BME-like I2C SDA / SCL	GPIO 21 / 22	Placeholder alternative; verify module supply, pull-ups to safe logic voltage, address and wiring length.
Power / ground	Board/module specific	Use a suitable protected supply, common reference as required by the interface, strain relief and dry enclosure; avoid unintended USB/external supply back-powering.

- ESP32 ADC inputs are **not 5 V tolerant**. “0–3.3 V conditioned output” is not a

promise of linear/full-scale ADC accuracy; confirm the actual board's permitted input and usable ADC range at the selected attenuation. Measure output limits before attaching the ADC. Add appropriate conditioning/protection under qualified hardware review; do not choose a resistor divider without considering probe loading.

- GPIO 34/35 on classic ESP32 boards are input-only and lack internal pull-ups;

other variants may differ. Avoid assuming the same pin map on every ESP32 family.

- The sketch sets 12-bit reads but does not validate reference voltage, attenuation,

ADC nonlinearity, grounding/noise, sensor excitation or temperature compensation. Do not treat `3.3/4095`, raw-count endpoints or pH slope constants as calibration.

- Keep electronics/connectors dry and separate from sample containers. Clean sensor

contact surfaces between samples using the manufacturer's procedure. Do not taste samples or deliberately inhale suspect material. Follow the qualified sampling plan for protective equipment, biological hazards and disposal.

Record the reviewed wiring revision, voltage measurements, board/module datasheets, sensor IDs and reviewer in the device inventory.
Hold a device out of service for damaged cables, unknown voltage limits, failed reference checks, contamination or expired calibration under the approved method.

3. ESP32-to-browser serial contract

- **115200 baud, 8 data bits, no parity, one stop bit, no flow control.**
- UTF-8 JSON: one object followed by LF; the browser accepts CRLF too. No boot

banners, debug text or pretty-printed multi-line objects in the capture stream.

- Template emits approximately once per second. The app sends **no commands** and

reads one frame per explicit capture. Port selection is included in the **15-second** window; maximum frame size **16,384 bytes**.

- Required keys: `device_id`, `captured_at`, `calibration_status`, `readings`.

Optional: `calibration_reference`. Unknown keys, duplicate object keys and duplicate reading fields are rejected; do not add a `demo` or `protocol_version` wire key.

- `device_id`: 1–120 characters; optional reference: 1–200; neither accepts surrounding

whitespace/control characters. Stable eFuse-derived `nutriscan-esp32-<hex>` is an identifier, **not authentication**.

- Timestamp: `YYYY-MM-DDTHH:mm:ss.fraction` (fraction up to six digits).

It must be an actual valid timezone-aware timestamp; the parser does not establish that a clock is fresh or trustworthy.

- Calibration: `unknown|reported_current|reported_expired`, a device claim rather

than app verification. Check the underlying calibration record independently.

Allowed `readings` array: **1–7 unique fields**, finite JSON numbers (not strings), exact units and bounds:

Fields	Unit	Inclusive range
`moisture_pct`, `humidity_pct`, `protein_pct`, `dry_matter_pct`, `fiber_pct`	`%`	0–100
`ph`	`pH`	0–14
`temperature_c`	`°C`	-20–80

The three visual scores cannot be captured over this protocol. An example frame below is **synthetic demo data**, with a fixed illustrative timestamp, not a field measurement. Keep the demo label in session records because the wire has no demo flag:

```
{"device_id":"host-emulator-01","captured_at":"2026-10-15T12:00:00+05:30","calibration_status":"unknown","readings":
[{"field":"moisture_pct","value":62.0,"unit":"%"},{"field":"ph","value":4.2,"unit":"pH"},
{"field":"temperature_c","value":24.0,"unit":"°C"},{"field":"humidity_pct","value":68.0,"unit":"%"}]}
```

4. Bring-up and capture procedure

1. Log device/mode/firmware digest; inspect wiring with power removed. For a demo,

keep `DEMO_MODE=true` and update the manual timestamp before compiling. For real operation, use a separately validated sensor/time implementation and reference checks.

1. Open the sketch in an existing Arduino IDE/ESP32 core environment, choose the

correct board and port, compile/upload and retain the actual build/upload result.
No successful firmware compile or physical wiring check is asserted by this pack.

1. Check serial output at 115200. Close the serial monitor before the browser opens

the port; only one process should own it. The README's host smoke only checks JSON and keys; it is not full protocol, calibration or electrical validation.

1. On Linux, an optional host transport check (substitute the actual port) is:

```
stty -F /dev/ttyUSB0 115200 cs8 -cstopb -parenb -ixon -ixoff raw -echo
timeout 10s python3 -c 'import json,sys; [print(json.dumps(json.loads(line))) for line in sys.stdin]' < /dev/ttyUSB0
```

A timeout exit status of 124 is expected when bounded streaming ends; inspect output/errors and valid frames, rather than calling that exit code a QA pass.
For no-hardware transport use the exact PTY emulator.

1. In a supported Chromium browser on HTTPS or localhost, sign in as owner/member,

enter screening step 3, open instrument capture and explicitly select the port. Review ID, actual time, calibration reference/state, values and units. Select readings and acknowledge review before confirming. No automatic scan is submitted.

1. Fill remaining actual measurements and perform the normal final review/submit.

Direct online saves retain selected-field provenance. Editing a captured value removes that field's provenance; changing source/device/time removes the capture. A new capture replaces previous capture provenance, not all previous form values.

1. Refresh history and confirm saved values/provenance. On timeout, unplug, malformed

frame or unsupported browser, record the failure and use measured manual entry; never fill absent values from the demo constants.

5. Device and sensor calibration records

Use inventory for board ID, asset tag, sensor/module serials, firmware and wiring revision, mode, owner, clock source, validated voltage limits and calibration references. Store actual sensor serials and multiple calibration IDs in the controlled `sensor_inventory_ref` record.

Use calibration-record.csv for **each reference point**: sensor, method/version, buffer/reference lot and certificate, sample/basis, environmental temperature, raw ADC count/voltage, as-found and as-left value, reference value/unit, tolerance, error (`as_left - reference`), result, due time, reviewer and raw evidence hash. Multiple points share a calibration record ID but have distinct point IDs. A failed as-found check can affect earlier samples even if adjustment makes as-left pass; assess that interval explicitly.

Choose calibration frequency and acceptance tolerance with the module supplier and qualified measurement lead; there is no app-enforced due-date or device registry. Keep sample-linked checks in the custody record and follow calibration/model release before using sensor results to substantiate accuracy claims.

Production Launch Checklist

Production launch and monitored handover checklist

Initial decision: not assessed / no approval recorded. This checklist does not convert the MVP into a production multi-tenant service. Distinguish a synthetic demo, a restricted pilot and broader production; approval applies only to the recorded scope. Every checked item needs named owner/date/evidence in the release record.

Release record — fill before review

Field	Required value
Release ID / candidate	PLACEHOLDER_RELEASE_ID; source revision plus working-tree/artifact digest
Scope / environment / URL	PLACEHOLDER_SCOPE; authorized farms/users/devices/sample types/languages; HTTPS origin
Backend / frontend / config	PLACEHOLDER_ARTIFACT_DIGESTS; redacted config revision; DB/schema pairing
Firmware / sensors / model	PLACEHOLDER_VERSIONS; demo/real status, calibration refs; current model is uncalibrated prototype
Cutover window / rollback artifact	PLACEHOLDER_TIME_WITH_TIMEZONE; previous known-good artifact/DB/backup refs
Named accountable people	PLACEHOLDER_RELEASE_AUTHORITY; operations, QA, hardware/calibration, data, domain, field, support and backups
Evidence location / support hours	PLACEHOLDER_RESTRICTED_EVIDENCE_REF; staffed hours and out-of-band escalation

Before the cutover window

Product, safety and field ownership — product/domain/field leads

- ☐ Readiness completed for the exact scope;

unresolved items and risk register have named dispositions.

- ☐ Prototype/lab-replacement limits verified in UI, exports, sharing, training and

launch script. No unsupported feed-safety, nutrient accuracy or contaminant claims.

- ☐ If releasing a calibrated model, model/data/lab gates

have approved numeric limits, locked real evaluation evidence and signed card.
Otherwise retain explicit uncalibrated status; synthetic metrics authorize no claims.

- ☐ Target hardware wiring, actual sensor hooks/time source, inventory and reference

checks approved. Demo units/data unmistakably separated from real pilot records.

- ☐ Operator SOP rehearsed; qualified adviser/provider contacts

and actual sampling/handling limits available; translation safety review completed.

- ☐ Local-only requests/reminders and public-only offline reload explained; no

promised lab booking, SMS, private offline history or draft provenance retention.

Deployment, identity and data — operations/data leads

- ☐ Candidate artifacts/configuration identified and staging rehearsal completed

under the deployment runbook; actual services/owners known.

- ☐ HTTPS/certificate renewal, DNS, root-path static assets and proxy forwarding

verified. API not directly public; `NUTRI_SCAN_COOKIE_SECURE=true` and exact origins set; deployed cookie login/reload/logout exercised on target browsers.

- ☐ Always-present dev origins, SameSite=Lax and server-own-origin behavior reviewed.

Any separate-origin topology proven; no wildcard CORS/private CDN cache.

- ☐ Registration exposure, access withdrawal, account recovery/reset and session

revocation procedures resolved for scope. Incomplete controls block broader launch or require explicitly approved restricted-pilot containment.

- ☐ Absolute durable SQLite path/permissions, disk/attachment headroom and measured

expected load acceptable. No assumption that more workers fix SQLite/throttling.

- ☐ Latest consistent backup verified, encrypted off-host copy available, restore

drill meets approved numeric RPO/RTO, sessions/deletion reconciliation rehearsed.

- ☐ Startup + first fieldwork lazy migration and any separate legacy assignment

verified on a copy; prior artifact/data compatibility or restore fallback proven.

- ☐ Governance retention/access/deletion schedule

approved with actual operators/jobs; local-device/export handling and backup expiry owned.

- ☐ Runtime secrets/TLS/backup credentials restricted; no secrets in frontend,

firmware, templates, logs or evidence; upload/PDF trust procedure confirmed.

QA and recovery — QA/support leads

- ☐ Required regression commands and acceptance cases executed

on the final candidate; actual results/outputs recorded, no historical counts reused.

- ☐ Cross-farm/role/Origin/private attachment/no-store gates pass. Viewer remains

read-only. All required input and idempotent retry paths pass.

- ☐ Saved-report/download/print, batch/history, observation-only, request/reminder,

explicit draft sync/provenance limits and field pilot smoke have evidence.

- ☐ Real-device serial permission/unplug/time/calibration and voice/manual fallback

tested for advertised devices; unavailable tests are blocked, not passed.

- ☐ PWA install/update/explicit activation/offline help/cache privacy and rollback

tested on the actual HTTPS host/browser matrix; coherent old hashed assets retained.

- ☐ Incident contacts/coverage, alert routing, rollback decision authority, restricted

evidence access and incident exercise confirmed.

Cutover — operations lead with release authority

1. Confirm go/no-go authorization and operator/support presence. Notify pilot users

of maintenance, explicit sync/update behavior and approved stop/manual fallback.

1. Pause writes at the edge, drain/stop the previous API, take the final verified

backup and record counts/checksum/time. Preserve the prior code/config/static artifact; do not overwrite the only rollback copy.

1. Deploy the complete candidate backend/static/worker/config pairing. Start under

the approved supervisor, initialize core tables, then exercise authenticated fieldwork to cover lazy schema creation. Do not auto-run legacy ownership assignment.

1. Run edge health/unauthenticated privacy probes and authorized scope-specific

read/write/attachment/report smoke. Verify DB readback, session flags, cache policy and sample/scan/provenance linkage. Delete/dispose of launch fixtures by policy.

1. Reopen only the authorized scope if checks pass. For unsafe claims, privacy/data

failures, corruption, broken core flows or unknown migration compatibility, keep service restricted and execute the rollback procedure.

Observation and support handover

Proposed initial observation: a **staffed first two hours** plus daily review for the first seven pilot days; approve coverage and actual numeric alert/stop thresholds in the release record before opening. These are planning defaults, not an SLA.

- ☐ Watch availability, request failures/latency, 401/403/429 patterns, disk/DB

growth/locks, backup age/failures, missing scans/queue conflicts, attachment errors, PWA update failures, sensor/reference drift and safety/privacy complaints.

- ☐ Confirm at least one real backup completes on schedule and assign the next

restore/access/retention review. A green `/health` is not sufficient evidence.

- ☐ Log deviations/incidents and next update times. Stop affected workflows for

cross-farm exposure, failed sensor checks, data loss or harmful interpretation.

- ☐ Hand over current release/DB/backup/rollback refs, device roster, sample/custody

gaps, unresolved queues, accepted exceptions, support contacts and next shift owner.

Final decision and approval

Record **go / restricted_go / no_go** with actual time, permitted scope, conditions, accepted risk IDs, expiry/next review, and named release-authority, operations, QA, data and domain approvals. Hardware/calibration approval is required for real-device claims. Missing critical safety/privacy/restore evidence means no real-data go-live. Publish only the approved scope; retain the decision and QA results in restricted evidence.

Qa Test Plan

QA test plan and acceptance criteria

Execution status: not run by this documentation pack. Prior results in fieldwork verification describe an earlier run; they do not certify this candidate, production HTTPS or physical sensors. QA lead records release/configuration/device versions and every execution in qa-test-results.csv.

1. Test policy and fixtures

- Test generated accounts and **clearly labelled synthetic demo data** against a

disposable SQLite DB, not the normal backend DB or real farm accounts. Use at least two farms, an owner/member/viewer, feed and silage, and repeated batch IDs across farms/sample types. No real external sharing, lab booking or SMS is needed.

- Run commands from the stated directories using existing dependencies. Browser

scripts additionally need an existing Playwright module, Chromium and runtime libraries; these are not installed by `npm test`. Record `blocked` if unavailable.

- Preserve stdout/stderr and exit status in restricted evidence. Include expected

versus actual behavior, issue reference and operator/time. Scrub passwords/cookies.

- `pass`: all assertions/expected outcomes met with evidence; `fail`: observed mismatch;

`blocked`: prerequisite missing; `not_run`: no execution. `not_applicable` needs a scope rationale and approver. Do not convert missing device testing into a pass.

2. Exact regression commands

From `backend/`:

```
.venv/bin/python -m unittest discover -s tests -p "test_*.py"
```

From `frontend/`:

```
npm test
VITE_PUBLIC_API_URL= VITE_API_URL= npm run build
```

From the repository root, for the independent synthetic calibration scaffold:

```
python3 -m unittest discover -s tools/calibration -p 'test_*.py' -v
```

Pass criteria: commands exit 0, no unexpected test failures/skips, build emits the production PWA files and no unresolved error/warning affecting scope. Record actual counts; do not expect historical 48/309 counts to remain fixed. Existing docs note a

FastAPI/Python deprecation warning; review the actual output. There is no configured separate lint/typecheck script. `git diff --check` from the root can check tracked whitespace; it does not check new untracked documentation by itself.

3. Isolated live browser setup

Ensure ports 8123/5174 are free; **do not stop existing user services to free them**. If occupied, arrange a separate test window/workspace. Live farmer/fieldwork scripts assert frontend port 5174; pointing them at arbitrary existing preview servers is unsafe because the scripts create test accounts/data.

From the root, create a fresh private disposable test location outside the repository:

```
ls -ld /tmp/omnirush
export SMOKE_DIR="$(mktemp -d /tmp/omnirush/nutriscan-launch-qa.XXXXXX)"
python3 -c 'import os; print(os.environ["SMOKE_DIR"])'
```

Use that **same printed path** as `SMOKE_DIR` in each terminal. Terminal A, from `backend/`, starts the disposable API:

```
NUTRI_SCAN_DB="$SMOKE_DIR/smoke.sqlite3" NUTRI_SCAN_COOKIE_SECURE=false \
.venv/bin/python -m uvicorn app.main:app --host 127.0.0.1 --port 8123
```

Terminal B, from `frontend/`, builds the app with no absolute API override and proxies to that API. A build replaces `dist/`; use a disposable test workspace if an existing build must be retained:

```
VITE_PUBLIC_API_URL= VITE_API_URL= npm run build
VITE_PUBLIC_API_URL= VITE_API_URL= VITE_PROXY_TARGET=http://127.0.0.1:8123 \
npm run preview -- --host 127.0.0.1 --port 5174 --strictPort
```

Terminal C, from `frontend/`, set an existing module path (omit that variable only if `playwright` already resolves), then run:

```
export NUTRISCAN_PLAYWRIGHT_MODULE=/PLACEHOLDER_EXISTING_INSTALL/node_modules/playwright
NUTRISCAN_SMOKE_URL=http://127.0.0.1:5174 NUTRISCAN_ALLOW_TEST_ACCOUNTS=1 \
node tests/farmer-flow.browser.mjs
NUTRISCAN_SMOKE_URL=http://127.0.0.1:5174 NUTRISCAN_ALLOW_TEST_ACCOUNTS=1 \
node tests/fieldwork-live.browser.mjs
NUTRISCAN_PREVIEW_URL=http://127.0.0.1:5174 node tests/pwa-live.browser.mjs
```

Optional isolated/mock browser scripts from `frontend/`, with the same Playwright prerequisite; they manage their own fixtures/servers:

```
node tests/fieldwork.browser.mjs
node tests/reading-assist.browser.mjs
node tests/pwa.browser.mjs
```

Pass: script exit 0 and expected assertions fulfilled; inspect actual output. Mocked serial/speech/share APIs do not prove permission prompts, physical transport, audio or external delivery. Stop only the test processes you started, retain needed redacted evidence and dispose of the isolated DB/profile under the retention policy.

4. Acceptance cases

Execute these against the candidate. Rows identify **required behavior**, not results. For API writes use the correct trusted `Origin` and authenticated session unless the test deliberately checks missing/foreign identity. Unknown/foreign object lookups should not leak another farm's existence.

ID	Exercise	Pass criteria
AUTH-01	Register, login, reload, logout; bad login	New farm/owner, password length >=10; HttpOnly session, no token/password in localStorage; logout revokes session; bad login 401; HTTPS cookie has Secure and SameSite=Lax
AUTH-02	Owner creates member/viewer; roles attempt writes	Owner manages members; member can scan/upload; viewer reads/downloads but scan/fieldwork/upload writes return 403; non-owner member creation denied
AUTH-03	Second farm, forged `farm_id`, guessed IDs, foreign attachments	No cross-farm read/write; explicit mismatched farm filter/body 403 where supported; missing/foreign private resources 404; no-session routes 401
AUTH-04	Missing/untrusted Origin on mutation, allowed Origin, repeated bad logins	Missing/untrusted Origin 403; legitimate origin succeeds; throttle returns 429 under its documented limit; all private responses/errors no-store/nosniff
SCAN-01	Blank, malformed, nonfinite and out-of-range inputs; actual zero	UI blocks missing/invalid values and requires all ten; zero remains zero; visual inputs 0–1; percent inputs 0–100, pH 0–14, UI/instrument temperature -20–80; no fabricated defaults
SCAN-02	Four steps, Back, edit after confirmation, submit/refresh	Back retains values; edits invalidate review; persisted result/IDs match reviewed request; prototype and uncalibrated metadata visible
SCAN-03	Repeat same farm/request ID with same/changed payload after restart	Equivalent request returns same saved scan, no duplicate; changed payload 409; scope remains farm-local
HIST-01	Search text/exact batch, UTC date bounds, filters and pagination	Literal matching; inclusive UTC dates; all expected farm records reachable; feed/silage not merged; missing trend values remain missing
REPORT-01	Export then edit unsaved form; refresh/export history; popup blocked	HTML represents saved record/ID only; print hides controls but keeps measurements/advisory/disclaimer; PDF saved via browser dialog; blocked popup explains HTML fallback
PHOTO-01	Valid JPEG/PNG/WebP; metadata-bearing, bad type, too large, sixth photo	Private evidence saved/readable by farm; normalized and metadata stripped; <=5 MiB and max five/parent; invalid payload/type/size/count rejected; no AI-image claim
SHARE-01	QR on another device; preview summary and selected evidence	QR contains batch route, no token or access grant; sign-in required; farm identity/photos off by default; text share has no private URLs/tokens; photos only through supported explicit native file sharing
DRAFT-01	Save/queue offline in open signed-in app; reconnect/retry; switch user	No implicit upload on reconnect; explicit sync idempotent; failures remain actionable; no other user's drafts visible; no photos/tokens/API responses stored
DRAFT-02	Save/load instrument-origin draft	Readings can persist but device/time/calibration provenance is stripped; warning visible; resulting record is not presented as retaining instrument provenance
FIELD-01	Create/edit batch; duplicate identity; Use this batch	Unique farm/batch/sample, immutable batch/sample identity; valid dates/quantity; reuse copies identity only, never old readings
FIELD-02	Save Yes/No/Unknown observations, including untouched answers	Unknown stays unknown; `observation_only`; no engine call, score, confidence or safety verdict
FIELD-03	Reminder complete/reopen; local date due groups	Correct persistence and calendar grouping; no promise of SMS/background delivery
FIELD-04	Request → collected → report_received, invalid backward/date paths	Forward transitions/date rules enforced; report required before received; tracker never says booked/sent automatically
FIELD-05	Request report upload/download; photo/report limits	Max five attachments; photos <=5 MiB, PDF <=10 MiB; basic PDF signature/EOF validation, attachment download/no inline execution; no OCR/malware-scan claim

ID	Exercise	Pass criteria
SERIAL-01	Valid 1–7 reading frame, select subset, cancel/edit/recapture	Explicit user port/review/application only, no commands/submit; unselected readings unchanged; provenance matches selected values; edits drop affected provenance
SERIAL-02	Unknown/duplicate keys/fields, numeric strings, wrong units/time, oversize, timeout/disconnect	Invalid frames not applied; no invented readings or scan; clear failure/manual fallback; 16,384-byte/15-second boundaries exercised
VOICE-01	Consent denied, stop/unmount, reviewed value, language unavailable	No automatic microphone use or submission; typed fallback always available; browser/provider/network dependence clear; app stores no audio
PWA-01	HTTPS production install, inspect caches, offline reload in three languages	Correct manifest/icons/root scope; only public assets/help cached; API/auth/photos/reports network-only; offline page exposes no farm data/login form
PWA-02	Candidate update, explicit activation and old-release rollback	No forced activation while unsaved; coherent worker/assets; old namespaces removed on activation; private cache boundary survives both versions
OPS-01	Backup, restore, startup + first fieldwork call, role/attachment readback	Integrity/FK checks pass, accounts/scans/attachments recover, sessions revoked in restore, deletion ledger reconciled; approved RPO/RTO met
CAL-01	Synthetic scaffold tests/report and label review	CLI report is explicitly `synthetic_demo_not_for_field_use`; coefficients not connected to app; no lab-accuracy claims
UX-01	390px and target devices; English/Hindi/Marathi; listening and keyboard	No hidden required control/overflow blocker; readable text/icons and focus; actual voices/translation approval recorded; manual fallbacks usable

The raw manual API `SensorData.temperature_c` currently accepts finite numbers without the UI/instrument -20–80 bound. Do not assert API parity for that field; record the difference and decide whether a broader-input API is acceptable for the release scope. Client validation is not a substitute for server validation.

5. Field pilot smoke — field lead + QA witness

Use authorized pilot accounts on the pilot environment and actual approved devices; label any demo fixtures synthetic. Preserve real and synthetic records separately.

1. Record site/date, target phone/browser/OS, network, release, device/sensor IDs,

calibration/time checks and operator training. Confirm support can be reached.

1. Sign in; create/select the correct batch/sample; capture one observation with

Unknown preserved and confirm it has no score.

1. Run reference checks, then an actual instrument capture (or measured manual

fallback), review all ten inputs and submit **only if complete**. Refresh saved history/provenance; compare readings against the operator record.

1. Download a saved HTML report and review the print/PDF flow. Test authenticated

QR access from another device. Review a share preview without sending externally.

1. On a disposable complete-reading draft, disconnect/reconnect and demonstrate

explicit sync, count/history confirmation and the provenance-loss warning. Test offline public help separately; do not expect private history after offline reload.

1. Demonstrate reminder complete/reopen and local testing-request/report tracking

with a clearly synthetic trusted report; never claim a lab was contacted.

1. Record serial unplug/permission denial, real speech/manual fallback, logout/shared

device handling, discrepancy/referral and end-of-shift reconciliation.

Pass: every in-scope step meets acceptance criteria, no unaccounted sample/scan/queue item, no misleading safety claim and no unresolved critical/high failure. Stop real-data capture for failed reference checks, unsafe hardware or farm isolation failures. Record unavailable conditions as blocked and schedule a witnessed rerun.

6. Exit and regression policy

Required regression commands and all scope-required acceptance cases must pass for the exact candidate. Critical safety/privacy/data-loss failures are no-go; high functional failures block affected workflows. Lower-priority exceptions need a named owner, bounded scope, workaround, expiry and release-authority approval. Device-specific features cannot be advertised as verified while their real-device checks are blocked.

On a fix, rerun the failing case and affected neighboring cases; rerun the broader required suites when the candidate changes. Attach QA summary to production launch, and log incidents/discrepancies under the support runbook.

Risk Register

Technical, product, safety and business risk register

Initial assessment on **2026-10-15**; all entries are **open / mitigation unverified**. Likelihood/impact are qualitative planning estimates (`L|M|H`), not measured rates. Named individuals, numeric thresholds, review dates and residual-risk acceptance belong in the restricted release record. Role titles below are unassigned owners until a person accepts them. No mitigation is considered completed by documenting it.

ID / category	Failure and consequence	Likelihood / impact	Mitigation and required evidence	Accountable role / review trigger
R01 safety/product	Prototype risk/confidence mistaken for lab accuracy or safe-to-feed approval	H / H	Prominent uncalibrated disclaimer in results/reports/shares/training; qualified referral; no certification/contaminant claims; wording review and field comprehension evidence	Domain/product lead; every claims/score/translation change
R02 hardware/safety	Raw pH probe, 5 V ADC signal, unsuitable supply or wet damaged wiring harms board/operator or corrupts readings	M / H	Exact-board datasheets, conditioned interface, powered-off inspection, measured voltage limits, protected enclosure and qualified wiring sign-off	Hardware lead; every module/wiring change or damage
R03 validity	Demo constants, placeholder ADC conversions, missing temperature/humidity hooks or stale manual time appear to be real observations	H / H	Label demo units/sessions; separate synthetic records; implement validated hooks/clock; inventory/calibration evidence and actual reference tests	Hardware/calibration lead; each device/firmware/session
R04 data/model	Unrepresentative samples, lab basis mismatch, contaminated material or split leakage produces optimistic accuracy	H / H	Custody and lab QC, immutable raw hashes, train-only transforms, farm/batch/time grouping, locked test and subgroup thresholds; independent release review	Calibration/data leads; dataset/model/method/domain change
R05 security/privacy	Cross-farm reads/writes or trusted-Origin/cookie misconfiguration exposes farm data	M / H	Farm-scoped auth/role/attachment tests, HTTPS/Secure cookie and exact origins; review always-on localhost/own origins and edge restrictions	Security/operations lead; deployment/auth/proxy change
R06 identity/business	Public account creation abuse, incomplete reset/removal workflow or process-local throttle enables unauthorized use/support overload	H / H	Restricted-pilot edge access, named account/admin process and incident contacts; resolve public registration/withdrawal/recovery controls before broad launch; abuse/load evidence	Product/security lead; exposure expansion or staff change
R07 operations/data	SQLite write contention, attachment growth, disk exhaustion or corruption loses scans/availability	M / H	Single-process starting topology, real load/capacity measurement, storage alerts, bounded uploads, consistent encrypted backups and restore drill; capacity-based DB migration decision	Operations lead; load/storage trend or backup failure
R08 release/data	Additive/lazy schema changes or legacy assignment break old code, misassign ownership or make rollback unsafe	M / H	Copy-based startup + first-fieldwork migration rehearsal, ownership review, before/after IDs/counts, compatibility pairing, pre-change backup, rollback drill	Release/operations lead; migration/assignment/release

ID / category	Failure and consequence	Likelihood / impact	Mitigation and required evidence	Accountable role / review trigger
R09 offline/data	User assumes private offline reload, auto-sync or preserved provenance; loses drafts/traceability or duplicates a scan	H / H	Public-help-only messaging, explicit sync/idempotency tests, draft provenance warning, protected independent capture log, end-of-shift reconciliation and no blind storage clearing	Field/frontend lead; queue/PWA/storage change
R10 PWA/privacy	Cache policy change/CDN caches private data, or mixed worker/assets strand old tabs	M / H	Allowlist cache inspection, private no-store/network-only tests, atomic artifact/worker update, retained old hashed assets, explicit activation and rollback test	Frontend/operations lead; build/cache/CDN update
R11 browser/product	Web Serial, microphone, speech language, native sharing, printing or install unsupported on target devices	H / M	Actual HTTPS phone/browser matrix, permission/unplug tests and usable measured manual/readable/download fallback; do not equate mocks with device evidence	QA/field lead; new device/browser/OS
R12 attachments/privacy	Trusted-looking PDF is malicious; photos/reports/exports/shared devices reveal identity	M / H	Basic PDF validation explicitly not scanning; trusted-provider handling, isolated review, photo metadata stripping, preview-first sharing, local/export deletion guidance and privacy incident drills	Data/support lead; new file flow or exposure
R13 UX/safety	Draft translations, unclear units/visual scores or inaccessible controls change an operator's decision	M / H	Expert English/Hindi/Marathi review, units/basis training, unknown stays unknown, no fabricated values, 390px/actual-device operator comprehension checks	Domain/field lead; copy/language/workflow change
R14 product/safety	Local testing request/reminder implies real lab booking, certified provider, report interpretation or SMS	H / H	Explicit local-tracker labels, manual provider/contact confirmation, report-received gating, qualified interpretation and SOP comprehension test	Product/field lead; testing/reminder integration change
R15 governance/business	Unexecuted retention/deletion, resurrected backup data, lost device or unowned support creates privacy/reputation cost	M / H	Approved schedule/owners, deletion ledger, restore reconciliation, access reviews, local-device cleanup and incident/on-call exercise	Data/operations lead; restore, withdrawal or lost device
R16 business/scale	Sensor/lab/logistics costs, support load, poor repeatability or low farmer adoption defeats viability	H / H	Bounded pilot records actual cost per usable sample, lab turnaround, repeat visits, support tickets and farmer comprehension/value; pre-agreed budget/service thresholds before expansion	Business/product lead; pilot review or budget breach
R17 contract/validity	Raw manual API accepts finite temperatures outside UI/serial -20–80 range; integrations assume identical validation	M / M	Document current contract, inventory clients, test UI/serial bounds separately and decide server-range policy before third-party API exposure; record exception/scope	Backend/product lead; external integration/schema change

Review and acceptance procedure

1. At readiness review, assign a named owner, target date, evidence reference, current/residual rating and proposed disposition (`mitigate|avoid|accept|block`).
1. Review weekly during a pilot, before each candidate release and immediately after any trigger/incident. Track actual likelihood evidence rather than reducing ratings because a checklist exists.
1. Safety/privacy/data-loss risks with missing critical controls block real-data operation. A synthetic-only demo can proceed only under that explicit scope; it does not accept the same risks for farmers or production data.
1. Release authority and the relevant domain/data owner approve bounded residual

risk with a reason, scope, expiry and stop trigger. No numeric validation gate may be waived by citing a synthetic scaffold result.

1. Escalate materialized risks through support/incidents

and feed corrective actions into QA,
readiness and
final launch.

Sources for current limitations: README,
fieldwork contract,
real-device gaps,
ESP32,
calibration workflow.

Support And Incident Runbook

Support and incident runbook

Before launch: replace `PLACEHOLDER_SUPPORT_ROUTE`, `PLACEHOLDER_ON_CALL_CONTACT`, `PLACEHOLDER_DATA_STEWARD` and `PLACEHOLDER_QUALIFIED_ADVISER` in the restricted contact sheet. Name primary/backup owners, actual coverage hours and escalation route. These placeholders are not working contact details or a 24/7 support promise.

Use incident-log.csv for incident summaries and a restricted linked timeline for updates. Do not place passwords, cookies, full private reports, raw farm contacts or identifiable screenshots in public tickets.

1. Intake and ordinary support

Collect: opaque ticket ID, user role/farm alias, time/timezone, release, page/action, browser/OS, online/offline state, error/status, reproduction steps and affected record IDs only where needed. Ask permission for minimized screenshots. Never ask the farmer to send a password/session token or run an unreviewed database command.

Symptom	First checks / resolution
Login returns 401/403	Check correct account, HTTPS/Secure cookie, origin/proxy configuration and system time; 403 may be Origin/role policy, not a bad password. No in-app password reset is supplied.
Login returns 429	Respect Retry-After; review throttling/proxy client-IP behavior. Do not repeatedly retry or disable controls as a quick fix.
Scan cannot submit	Confirm all ten actual inputs, valid units/bounds and final review; member/owner role. If measurements are unavailable, use observation-only tracking rather than demo fillers.
Queue appears stuck	Confirm correct user/farm, connectivity and explicit Sync now. Check saved history/request ID before retry; unchanged requests are idempotent. A 409 needs review, not a new ID to force a duplicate.
Device details missing after draft load	Expected allowlist limitation: readings retained, instrument provenance stripped. Use independent field records; do not claim the original device timestamp survived.
Serial/voice unavailable	HTTPS/localhost, supported browser, permission, correct port/115200, close serial monitor, inspect timeout/frame. Keep manual measured entry available. Speech may depend on an external browser service/network.
Cannot upload evidence	Check supported type, <=5 MiB photo/<=10 MiB PDF, max five per parent, valid file and writer role. PDF checks are basic; do not open untrusted reports.
Print/install fails	Permit report popup or download HTML then print. Install depends on browser; test built root-path PWA over HTTPS. Offline reload is public help only.
Missing testing/reminder notification	Requests are farm-local tracking, not lab bookings. Reminders are in-app, not SMS/background notifications. Confirm manually with the selected provider.
Conflicting score/lab result or suspect feed	Stop relying on the score for that batch, preserve IDs/measurements, refer to a qualified lab/adviser. Scores/confidence are uncalibrated and never authorize feeding.

2. Severity and response targets

Targets below apply **during the explicitly staffed coverage window** and must be approved before launch. Record actual acknowledgment times; never backfill a target as an achieved response.

Severity	Examples	Target acknowledgment / coordination
SEV1 critical	Cross-farm exposure, active credential/backup compromise, suspected harmful safety claim/use, widespread corruption/data loss	15 minutes; incident lead + operations + data/domain lead immediately; stop affected real-data use
SEV2 high	Core login/scan unavailable, failed restores/backups threatening RPO, major wrong result/provenance, failed calibration affecting collected samples	1 hour; incident lead coordinates same-window containment and qualified review
SEV3 medium	One optional feature/device failing with safe manual fallback; isolated report/UI defect	1 business day; support/feature owner tracks workaround and fix
SEV4 low	Cosmetic/documentation issue without behavior or safety impact	3 business days; backlog with owner and next review

Outside coverage, document the actual escalation channel and safe stop/manual fallback instructions. Upgrade severity if scope is unknown and safety/privacy could be affected; incident lead records the reasoning and later reclassification.

3. Response sequence

1. **Declare / assign.** Create incident ID, severity, detection time, affected

environment/version, reporter alias, incident lead, operations owner, data/domain reviewer and next update time. Start a timezone-aware timeline.

1. **Contain.** Pause affected capture/writes/exports or restrict the edge; quarantine

affected hardware/sample batches. For private cache exposure, prevent further distribution and correct worker/static policy under controlled recovery. For potential feed harm, tell operators to obtain qualified advice, not to use a higher/lower prototype score as a safety test.

1. **Preserve evidence.** Keep a protected consistent DB snapshot, artifact/config

digests, redacted logs, request/scan/sample/device IDs, clock/calibration status, error counts and minimal screenshots. Follow custody/access controls. Do not delete or alter the only record of what happened.

1. **Assess.** Determine start/end time, farms/users/records, app/firmware/model

version, data categories, potential recipients, affected calibration interval and whether backup/deletion obligations are involved. Separate observed facts from hypotheses and list unknowns.

1. **Recover.** Fix/redeploy or use the rollback runbook.

Do not change DB schema, replay queues or downgrade code against an unknown schema ad hoc. A DB restore needs session revocation and deletion reconciliation.

1. **Verify.** Reproduce the failed path in isolation; rerun scoped

QA acceptance, isolation/attachment/readback checks and the affected real-device/reference checks. Observe service/queue/error metrics for an incident-lead-approved period before reopening.

1. **Communicate / close.** Report impact, containment, user action, remaining uncertainty and next update. Data steward coordinates privacy communications; domain lead approves safety wording. Record actual recovery/data loss and obtain closure approval. Schedule corrective actions, owners and due dates.

4. Operator diagnostics

Use actual environment paths/hostnames. These read-only probes do not authenticate or prove DB health. Expected unauthenticated private status is 401, not 200:

```
export APP_ORIGIN=https://nutriscan.example.invalid
curl --fail --show-error --silent "$APP_ORIGIN/health"
curl --show-error --silent -D - -o /dev/null "$APP_ORIGIN/api/v1/auth/me"
curl --fail --show-error --silent -I "$APP_ORIGIN/sw.js"
```

Authorized host operators can check a protected DB **read-only**; results may be resource-intensive on a large DB, so schedule appropriately. This does not migrate or create tables. From any directory, with the correct absolute path:

```
export NUTRI_SCAN_DB=/srv/nutriscan-data/PLACEHOLDER_PRODUCTION.sqlite3
python3 - <<'PY'
import os, sqlite3
from pathlib import Path
path = Path(os.environ['NUTRI_SCAN_DB']).resolve()
assert path.is_file()
db = sqlite3.connect(path.as_uri() + '?mode=ro', uri=True)
try:
    print('integrity:', db.execute('PRAGMA integrity_check').fetchall())
    print('foreign_key_violations:', len(db.execute('PRAGMA foreign_key_check').fetchall()))
finally:
    db.close()
PY
```

Record output in restricted evidence. A green health endpoint with a failed DB check is not recovery. Inspect disk/backup status through the host's existing monitoring, and confirm an authenticated scan/fieldwork read on an authorized test scope without logging response bodies or secrets.

Emergency session revocation

This is a **privileged, global write**: it signs out every user, not just the reporter. Incident lead approval, edge restriction, stopped/drained API, verified backup and confirmed environment are prerequisites. With the correct `NUTRI_SCAN_DB`:

```
python3 - <<'PY'
import os, sqlite3
from pathlib import Path
path = Path(os.environ['NUTRI_SCAN_DB']).resolve()
assert path.is_file()
db = sqlite3.connect(path.as_uri() + '?mode=rw', uri=True)
try:
    with db:
        revoked = db.execute('DELETE FROM sessions').rowcount
    print('Sessions revoked:', revoked)
finally:
    db.close()
PY
```

This does **not** disable accounts, reset passwords or prevent a compromised credential from logging in again. Keep affected access restricted until an approved identity recovery/removal process resolves the compromise. Rotate compromised host, TLS/backup or future integration credentials using the responsible secret store; there is no shared application session-signing secret to rotate in this MVP.

5. Communication, post-incident review and handover

Use this redacted update structure:

> Incident PLACEHOLDER_ID, detected PLACEHOLDER_TIME: observed impact and affected workflow/scope. Containment in place: PLACEHOLDER_ACTION. Operator action: PLACEHOLDER_ACTION. Known unknowns: PLACEHOLDER_UNKNOWNS. Next update: PLACEHOLDER_TIME via the approved support route. Prototype outputs remain screening indicators; seek qualified advice for feed/silage safety concerns.

For SEV1/SEV2 propose a review within two business days after stabilization. Capture timeline, cause/contributing conditions, detection/response gaps, actual data loss, sample/model implications, backup/restore results, deletion reconciliation, affected user follow-up, failed controls and concrete prevention actions with owners/dates.
Update the risk register, SOP/training and relevant QA cases.
Close only after evidence is reviewed, access is safe, affected samples are dispositioned by the domain lead and residual work is owned. Hand over open incidents explicitly at every shift change; silence is not a completed handover.

Technical Readiness Checklist

Technical readiness checklist

Status: unassessed. Every item below needs an owner, date, evidence reference and decision in the release record. Use QA results for executed checks. Missing evidence is `not_run` or `blocked`, never a pass.

1. Scope and ownership — release lead

- ☐ Select scope: synthetic demo, restricted real-data pilot, or broader production.

State farms, users, languages, sample types, devices and permitted claims.

- ☐ Assign release, operations, QA, hardware, calibration, data, field and support

leads plus backups and an out-of-band contact route. Fill actual support hours.

- ☐ Record the exact build/source/configuration and approved rollback artifacts.

Preserve unrelated working-tree work; do not deploy an unidentified dirty tree.

- ☐ Confirm the uncalibrated-prototype statement is visible in onboarding, saved

results, reports, share previews, demo scripts and operator training.

- ☐ Explicitly exclude lab equivalence, certified contaminants and automated photo

diagnosis until the calibration gates are met.

- ☐ Review risks; document residual risk owners and expiry dates.

2. Hosting and persistence — operations lead

- ☐ Rehearse deployment in an isolated staging database.

Dev Compose and Vite preview are not production process management.

- ☐ Use HTTPS with a valid certificate, root-path PWA hosting, restricted backend

network access and explicit trusted reverse-proxy settings.

- ☐ Verify `NUTRI_SCAN_COOKIE_SECURE=true`, allowed frontend origins and actual

cookie behavior in deployed browsers. No wildcard credentialed CORS.

- ☐ Review always-added localhost origins, server-own-origin acceptance and

SameSite=Lax constraints; edge controls and topology must match the approved scope.

- ☐ Use an absolute `NUTRI_SCAN_DB` on durable local storage with restricted access;

confirm free space, attachment growth, SQLite writer load and restart persistence.

- ☐ Complete consistent encrypted backups, checksum/integrity checks, restore

drill and migration rehearsal including first fieldwork access (lazy table creation).

- ☐ Approve numeric RPO/RTO and backup schedule. Proposed defaults in

governance require actual operational evidence.

- ☐ Provide an owned service supervisor, TLS renewal, monitoring and alert routing.

`/health` alone does not prove DB readability, capacity or authenticated writes.

3. Identity and privacy — data/security lead

- ☐ Verify owner/member writes, viewer read-only, cross-farm isolation, trusted

Origin checks and private `no-store` responses, including errors/attachments.

- ☐ Confirm there is no default account or embedded credential. Keep runtime

configuration and protected TLS/backup credentials out of builds and logs.

- ☐ Resolve public registration, account withdrawal, password recovery and session

revocation for the chosen scope. These administrative workflows are incomplete
in the MVP; use a documented restricted-pilot procedure or block broader launch.

- ☐ Approve the retention/access/deletion policy, local draft/shared-device handling,

sharing education and restore-time deletion reconciliation.

- ☐ Assess trusted PDF handling (basic validation is not malware scanning), logging

redaction and optional browser speech-provider processing.

4. Hardware and data validity — hardware/calibration leads

- ☐ Complete wiring review against the exact

ESP32 board/module datasheets; no raw pH probe or 5 V ADC connection.

- ☐ Record inventory, firmware digest,

pins, voltage limits, time source and sensor IDs for each device.

- ☐ Mark demo units and synthetic data unmistakably. Do not treat `DEMO_MODE=false`

as validated; moisture/pH conversions are placeholders and temperature/humidity
hooks need implementation before real measurement.

- ☐ Record reference checks, calibration intervals/tolerances and signed results

using calibration records.

- ☐ Test real port permission, unplug/reconnect, invalid frames, timeout, partial

readings and reviewed-only application on actual target devices.

- ☐ Establish sample/lab chain of custody and immutable raw records for any

accuracy evaluation. Synthetic scaffold metrics cannot approve field use.

5. Product and field operation — QA/field leads

- ☐ Execute regression and acceptance tests for the candidate;

retain outputs and discrepancies without importing historical pass counts.

- ☐ Verify all ten inputs, saved-record reports, batch/sample separation, photos, QR authentication and explicit idempotent offline sync.
- ☐ Verify observation-only records have no score; requests do not imply lab booking; reminders have no SMS/background delivery promise.
- ☐ Verify built PWA update/rollback, public-only Cache Storage and offline help in English/Hindi/Marathi. Private responses must remain network-only.
- ☐ Demonstrate the loss of device/time/calibration provenance through draft save/load and define a traceable manual fallback for field sessions.
- ☐ Obtain qualified review of translated safety guidance and target-device readability/voice behavior; mocked browser speech does not prove a real microphone.
- ☐ Train operators with the SOP, escalation script and sample handling plan. Unknown observations and absent measurements stay unknown.

6. Decision

The release authority records `go|restricted_go|no_go`, permitted scope, blocked items and next review date. Safety, farm isolation, durable data/restore and required QA failures are **no-go** for real-data operation. An approved synthetic demo does not approve a field pilot. A pilot does not approve laboratory or safety claims. Complete the production checklist for the actual cutover and handover.

Source baseline: README, fieldwork contract, verification/gaps, ESP32.

Calibration Record

data_status	calibration_record_id	point_id	device_id	sensor_id	sensor_type	firmware_version	wiring_revision	calibrated_at	operator_id	method_ref	reference_type	reference_lot	reference_certificate_ref	sample_id	measurement_basis	temperature_c	raw_adc_counts	raw_voltage_v	as_found_value	reference_value	as_left_value	unit	tolerance_abs	error_as_left_minus_reference	result	due_at	reviewer_id	reviewed_at	raw_evidence_ref	raw_evidence_sha256	notes
placeholder_not_for_use	PLACEHOLDER_CALIBRATION_ID	PLACEHOLDER_POINT_ID	PLACEHOLDER_DEVICE_ID	PLACEHOLDER_SENSOR_ID	PLACEHOLDER_SENSOR_TYPE	PLACEHOLDER_FIRMWARE_VERSION	PLACEHOLDER_WIRING_REVISION		PLACEHOLDER_OPERATOR_ID	PLACEHOLDER_METHOD_REF	PLACEHOLDER_REFERENCE_TYPE	PLACEHOLDER_REFERENCE_LOT	PLACEHOLDER_CERTIFICATE_REF	PLACEHOLDER_SAMPLE_ID	PLACEHOLDER_WET_OR_DRY_BASIS							PLACEHOLDER_UNIT			pending		PLACEHOLDER_REVIEWER_ID		PLACEHOLDER_RESTRICTED_EVIDENCE_REF		Template row only - remove before operational use; blank values are not measurements

Device Inventory

data_status	device_id	asset_tag	board_model	board_serial	sensor_inventory_ref	firmware_version	firmware_sha256	wiring_revision	mode	ph_adc_pin	moisture_adc_pin	dht_data_pin	bme_sda_pin	bme_scl_pin	validated_adc_max_v	clock_source	calibration_record_refs	calibration_due_at	custodian_id	assigned_farm_id	status	last_checked_at	wiring_evidence_ref	notes
placeholder_not_for_use	PLACEHOLDER_DEVICE_ID	PLACEHOLDER_ASSET_TAG	PLACEHOLDER_BOARD_MODEL	PLACEHOLDER_BOARD_SERIAL	PLACEHOLDER_SENSOR_INVENTORY_REF	PLACEHOLDER_FIRMWARE_VERSION		PLACEHOLDER_WIRING_REVISION	demo							PLACEHOLDER_TIME_SOURCE	PLACEHOLDER_CALIBRATION_REFS		PLACEHOLDER_CUSTODIAN_ID	PLACEHOLDER_FARM_ID	not_in_service		PLACEHOLDER_WIRING_EVIDENCE_REF	Template row only - no device or voltage limit has been validated

Incident Log

data_status	incident_id	detected_at	reported_by_id	environment	release_id	severity	status	summary	affected_scope_ref	incident_lead_id	containment_at	containment_summary	restored_at	data_loss_assessment_ref	evidence_ref	evidence_sha256	next_update_at	closed_at	reviewer_id	follow_up_ref	notes
placeholder_not_for_use	PLACEHOLDER_INCIDENT_ID		PLACEHOLDER_REPORTER_ID	PLACEHOLDER_ENVIRONMENT	PLACEHOLDER_RELEASE_ID		open	PLACEHOLDER_FACTUAL_SUMMARY	PLACEHOLDER_AFFECTED_SCOPE_REF	PLACEHOLDER_INCIDENT_LEAD_ID				PLACEHOLDER_DATA_LOSS_ASSESSMENT_REF	PLACEHOLDER_RESTRICTED_EVIDENCE_REF				PLACEHOLDER_REVIEWER_ID	PLACEHOLDER_FOLLOW_UP_REF	Template row only - no actual incident or response time is asserted

Qa Test Results

data_status	run_id	test_case_id	release_id	environment	executed_at	executor_id	device_browser_ref	status	expected_result	actual_result	evidence_ref	evidence_sha256	issue_ref	reviewer_id	notes
placeholder_not_for_use	PLACEHOLDER_RUN_ID	PLACEHOLDER_TEST_CASE_ID	PLACEHOLDER_RELEASE_ID	PLACEHOLDER_ENVIRONMENT		PLACEHOLDER_EXECUTOR_ID	PLACEHOLDER_DEVICE_BROWSER_REF	not_run	PLACEHOLDER_ACCEPTANCE_CRITERION		PLACEHOLDER_RESTRICTED_EVIDENCE_REF		PLACEHOLDER_ISSUE_REF	PLACEHOLDER_REVIEWER_ID	Template row only - no test execution or pass is asserted

Sample Chain Of Custody

data_status	sample_id	aliquot_id	event_id	event_type	event_at	farm_id	batch_id	sample_type	sampling_plan_ref	operator_id	from_custodian_id	to_custodian_id	device_id	calibration_record_id	container_id	seal_condition	sample_condition	temperature_c	handling_notes	lab_sample_id	lab_method_ref	lab_report_ref	raw_evidence_ref	raw_evidence_sha256	deviation_ref	notes
placeholder_not_for_use	PLACEHOLDER_SAMPLE_ID	PLACEHOLDER_ALIQUOT_ID	PLACEHOLDER_EVENT_ID	collection		PLACEHOLDER_FARM_ID	PLACEHOLDER_BATCH_ID		PLACEHOLDER_SAMPLING_PLAN_REF	PLACEHOLDER_OPERATOR_ID	PLACEHOLDER_FROM_CUSTODIAN_ID	PLACEHOLDER_TO_CUSTODIAN_ID	PLACEHOLDER_DEVICE_ID	PLACEHOLDER_CALIBRATION_ID	PLACEHOLDER_CONTAINER_ID	PLACEHOLDER_SEAL_CONDITION	PLACEHOLDER_SAMPLE_CONDITION		PLACEHOLDER_HANDLING_NOTES	PLACEHOLDER_LAB_SAMPLE_ID	PLACEHOLDER_LAB_METHOD_REF	PLACEHOLDER_LAB_REPORT_REF	PLACEHOLDER_RESTRICTED_EVIDENCE_REF		PLACEHOLDER_DEVIATION_REF	Template row only - no real sample or custody event is asserted